№1

σ-конечная мера - мера, при которой всё пространство можно покрыть счётным объединением множеств конечной меры.

$$E_t = \{x : f(x) > t\},\ t \geq 0$$

$$E = \{(x,t) : f(x) > t\} \subset X \times [0, +\infty) \Rightarrow \mu \otimes dt \text{ - мера на } E$$

$$\varphi(f(x)) = \int_0^{f(x)} \varphi'(t)dt = \int_0^{+\infty} I_{\{f(x)>t\}}\varphi'(t)dt$$

$$\int_X \varphi(f(x))d\mu = \int_X \int_0^{+\infty} I_{\{f(x)>t\}}\varphi'(t)dtd\mu = \int_0^{+\infty} \varphi'(t)\left(\int_X I_{\{f(x)>t\}}d\mu\right)dt$$

$$= \int_0^{+\infty} \varphi'(t)\mu\{x : f(x) > t\}dt$$

№2

a)

$$\forall p \geq 1 \text{ и } f \in L^p : \|f\|_p = \left(\int_X |f|^p d\mu\right)^{\frac{1}{p}}$$

1) $s < \infty$

$$(!) : \left(\int_X |f|^s d\mu\right)^{\frac{1}{s}} \leq \left(\int_X |f|^p d\mu\right)^{\frac{1}{p}}$$

$$\int_X |f|^s d\mu \leq \left(\int_X |f|^p d\mu\right)^{\frac{s}{p}}$$

$\left(\int_X |f|^s d\mu\right)^{\frac{p}{s}} \leq \int_X |f|^p d\mu$ по Гёльдеру $\Rightarrow$ (!) выполняется

$$\Rightarrow \|f\|_s \leq \|f\|_p \Rightarrow L_s \subset L_p$$

2) $s = \infty \Rightarrow f \in L^\infty$

$$|f(x)| \leq \|f\|_\infty$$

$$\Rightarrow \int_X |f(x)|^p d\mu \leq \int_X \|f\|_\infty^p d\mu = \|f\|_\infty^p \mu(X) = \|f\|_\infty^p$$

$$\Rightarrow \|f\|_p \leq \|f\|_\infty \ \forall p$$

б) Пусть $I(p) = \int_X |f|^p d\mu \Rightarrow \|f\|_p = I(p)^{\frac{1}{p}}$

$$\ln \|f\|_p = \frac{1}{p} \ln I(p)$$

$$\lim_{p\to 0} \ln \|f\|_p = \lim_{p\to 0} \frac{\ln I(p)}{p} = \left[\frac{0}{0}\right] = \lim_{p\to 0} \frac{I'(p)}{1} = \lim_{p\to 0} I'(p)$$

$$I'(p) = \frac{d}{dp} \int_X |f|^p d\mu = \int_X \frac{\partial}{\partial p} |f|^p d\mu = \int_X |f|^p \ln |f| d\mu$$

$$\lim_{p\to 0} I'(p) = \int_X \ln |f| d\mu$$

$$\Rightarrow \lim_{p\to 0} \|f\|_p = \exp\left(\int_X \ln |f| d\mu\right)$$

№3

по нер-ву Коши-Шварца:

$$\left(\int_X \sqrt{fg} d\mu\right)^2 \leq \left(\int_X f d\mu\right) \left(\int_X g d\mu\right)$$

$$fg \geq 1 \Rightarrow \sqrt{fg} \geq 1 \Rightarrow \int_X \sqrt{fg} d\mu \geq \int_X 1 d\mu = \mu(X) = 1 \Rightarrow$$

$$\Rightarrow \left(\int_X \sqrt{fg} d\mu\right)^2 \geq 1 \Rightarrow \left(\int_X f d\mu\right) \left(\int_X g d\mu\right) \geq 1.$$

№4

$$a = \mu(E) > 0$$

$$\int_X f d\mu = 1 \Rightarrow \int_E f d\mu \leq 1$$

$$\ln t \leq at - \ln a - 1$$

$$\ln f(x) \leq a f(x) - \ln a - 1$$

$$\int_E \ln f(x) d\mu \leq a \int_E f(x) d\mu - a \ln a - a \leq a - a \ln a - a =$$

$$= -a \ln a = -\mu(E) \ln \mu(E)$$